Voodoo Dough Boys

The Calculatorinator

User's Manual

Version 1.0

Revision History

Date	Version	Description	Author
01/05/26	1.0	Add to sections 1 to 6	Joseph Ubanda
02/05/26	1.0	Add to sections 7 and 8	Tram Dinh
05/05/26	1.0	Added to the introduction and how to download the program	Vu Nguyen
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1. Purpose

This User’s Manual serves as a guide for operating The Calculatorinator, providing instructions on how to input expressions, utilize supported operators, and interpret the software's output. The User’s Manual also provides a guide on installing the program on your own machine, as well as some common problems that the user may encounter.

2. Introduction

The Calculatorinator is an arithmetic expression elevator. It allows users to input mathematical expressions via a command-line interface and receive an accurate final evaluation of the expression.

Features:

- Addition, subtraction, multiplication, division
- Modulo operator
- Exponentiation
- Parentheses support
- Unary plus and minus
- Error handling for invalid input

3. Getting started

1. On the GitHub page(https://github.com/maddytd/The-Calculatorinator), press the Green Code button and copy the URL (https://github.com/maddytd/The-Calculatorinator.git).
2. Open a terminal and type in the command: **git clone https://github.com/maddytd/The-Calculatorinator.git**
3. Navigate to the src folder: ***cd ./The-Calculatorinator/src***
4. Compile the program using the command: ***make***
5. Open the program by using the command: ***./main***
6. When prompted, type your mathematical expression directly into the command line. Use the following operator symbols in your expression:
   a. +, addition
   b. -, subtraction
   c. *, multiplication
   d. /, division
   e. %,modulo
   f. **, Exponents
   g. and (), parentheses.
7. Press Enter to process the expression. The program will display the calculated result or an error message based on your expression.
8. To exit the program, enter ‘q’ or ‘Q’ as your input expression to exit the program.
9. Then run the following command to clean up the src folder: ***make clean***

4. Advanced features

The program follows the PEMDAS order of operations for evaluating the input expression. The program also supports decimal values within the expression.

5. Troubleshooting

Error:

"Division by zero" or "Modulo by zero."

- Within your expression, a value is being divided or modulo by zero

"Invalid float value."

- A float value has more than one decimal point. Or a floating-point value has a decimal point without a number after or before it. (Example: .5, 5.)

"Input cannot be empty."

- The input expression was empty

"Invalid character found."

- The input expression contains an invalid character that can't be evaluated

"Operator cannot be followed by a closing parentheses."

- The input expression has the inside of a parenthesis starting with a multiplication, division, modulo, or exponent operator

“Operator cannot follow opening parenthesis."

- The input expression has the inside of a parenthesis ending with an operator

"Empty parentheses."

- The input expression contained empty parentheses without any values

"Unmatched parentheses."

- The input contained unmatched parentheses within the expression

"Expression cannot start with an operator."

- The input began with a multiplication, division, modulo, or exponent operator

"Expression cannot end with an operator."

- The input ended with an operator

"Missing operator before parentheses."

- The input had a value and parentheses separated by no operator

"Invalid operator sequence."

- The input expression had two multiplication, division, modulo, or exponent operators with no value in between

6. Examples

----- WELCOME TO THE CALCULATORINATOR -----
(Powered by C++)
Please enter your algorithm to calculate below
(If you want to exit/quit the program type 'q')

Addition:
3 + 4
Final Result: 7

Subtraction with parentheses:
8 - (5 - 2)
Final Result: 5

Multiplication and Division:
10 * 2 / 5
Final Result: 4

Exponentiation:
2 ** 3
Final Result: 8

Mixed Operator:
4 * (3 + 2) % 7 - 1
Final Result: 5

Float values:
4.5 + 2.3
Final Result: 6.8

EXIT:
q
Thank you for using the Calculatorinator!

7. Glossary of terms

No special technical terms are required for this project. Common mathematical and programming terms are used throughout this manual.

8. FAQ

1. Does the program support decimals?

Yes, decimal values are supported.

2. Can I use spaces?

Yes.

Example: 3 + 4

3. Can I enter multiple expressions at once?

No. Enter one expression each time the program runs.

4. Why do I get an error?

Check for invalid characters, missing operators, or unmatched parentheses.

5. Does the program support values using scientific notation? (Ex. 2E+2 or 2e-2)

No. Our program doesn't support these values. These values will result in an "Invalid character found" error.